DOCUMENT 07 · INPUT AND OUTPUT CONTRACTS

Input and output contracts

The mandatory structure of every SPAD artefact, readable by people and by machines, so that the "incomplete artefact" validation can be automated and agents can chain phases without losing information

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18	1	3	4
contracts	block common to all	verdict values	severity levels

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Version under review: please do not circulate. The current state of SPAD (version 0.x) is not meant to be shared widely. It is public so that a small number of people can review it, give feedback and help improve it. Documents and tools are being adapted to make them reusable; this notice will disappear when the framework reaches version 1.x.

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01

Purpose

Each SPAD phase produces an artefact with a **contractual structure**: mandatory sections and fields that the instruction requires (document 06) and whose absence invalidates the response (document 03). This document defines that structure in JSON, so that:

- a person can check at a glance that the output is complete;
- an automatic validator can reject an incomplete artefact before anyone sees it;
- an agent that chains phases receives from the previous one exactly what it needs.

The artefact may be presented in Markdown with the same sections; JSON is the canonical form for validation and exchange.

> **Why it matters.** "Incomplete artefact" is the most frequent cause of invalidation and the easiest to automate. An explicit contract turns that check into a mechanical step and leaves to the person the judgement that does require a person: whether the content is correct.

02

Common conventions

2.1 Common block

Every artefact includes these fields:

```
{
  "spad": { "contract_version": "0.1" },
  "topic": "string, snake_case, maximum 20 characters",
  "phase": "PLAN | AUDIT_PLAN | CODE_PRIMER | TEST_STRATEGY | IMPLEMENTATION | TEST_IMPLEMENT
  "iteration": "integer from 1",
  "date": "ISO 8601",
  "model": {
    "provider": "string",
    "model": "string",
    "model_version": "string",
    "role": "planner | reviewer | builder | fixer | analyst | diagnostic | security_reviewer"
    "instruction": "identifier and version of the instruction used",
    "global_context": "version",
    "project_context": "version"
  },
  "deviations": [
    { "situation": "string", "decision_requested": "string" }
  ]
}
```

`deviations` records the situations that the phase could not resolve and on which a decision is requested; it never contains decisions taken by the AI.

2.2 Scales

Scale	Values
Verdict	`GO` · `GO_WITH_CHANGES` · `NO_GO` (all reviews, including the security review and the behaviour evaluation).
Severity	`CRITICAL` · `HIGH` · `MEDIUM` · `LOW`.
Probability	`HIGH` · `MEDIUM` · `LOW`.
Recommendation priority	`MUST` · `SHOULD` · `NICE_TO_HAVE`.
Quality	`EXCELLENT` · `GOOD` · `ADEQUATE` · `POOR` · `MISSING`.

2.3 Human validation record

It is not part of the AI's artefact: the person writes it in the topic's `VALIDATION.md` (document 02, section 4.2). Its canonical form:

```
{
  "artefact": "SPAD_02_AUDIT_PLAN",
  "person": "identifier",
  "date": "ISO 8601",
  "outcome": "VALID | INVALID | EXCEPTION_APPROVED",
  "invalidation_cause": "PHASE_VIOLATION | INCOMPLETE_ARTEFACT | OFF_PLAN | NO_PRIOR_REVIEW |
  "ai_verdict": "GO | GO_WITH_CHANGES | NO_GO | null",
  "verdict_accepted": "true | false | null",
  "observations": "string"
}
```

03

Main cycle contracts

3.1 PLAN

```
{
  "architecture": { "overview": "string", "patterns": ["string"] },
  "components": [
    { "name": "string", "responsibility": "string", "dependencies": ["string"], "interfaces":
```

```
  ],
  "flows": [
    { "name": "string", "trigger": "string", "steps": ["string"], "data": "string" }
  ],
  "decisions": [
    { "decision": "string", "justification": "string", "alternatives": ["string"], "trade_off
  ],
  "risks": [
    { "risk": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW", "probability": "HIGH|MEDIUM|L
  ],
  "ai_system": {
    "applies": "true | false",
    "autonomy_level": "A0 | A1 | A2 | A3",
    "ai_components": ["model, instructions, retrieval, tools, memory"],
    "action_limits": ["string"],
    "data_to_model": { "provider": "string", "information_classes": ["string"] },
    "failure_modes": [ { "failure": "string", "safe_behaviour": "string" } ],
    "model_change": "strategy for provider updates"
  }
}
```

3.2 AUDIT_PLAN

```
{
  "issues": [
    { "id": "string", "category": "COUPLING|COHESION|SCALABILITY|CONCURRENCY|SECURITY|OBSERVA
      "severity": "CRITICAL|HIGH|MEDIUM|LOW", "description": "string", "location": "section c
  ],
  "recommendations": [
    { "issue": "id", "recommendation": "string", "priority": "MUST|SHOULD|NICE_TO_HAVE" }
  ],
  "exceptions_evaluated": [
    { "global_rule": "string", "exception": "string", "evaluation": "string" }
  ],
  "verdict": "GO|GO_WITH_CHANGES|NO_GO",
  "justification": "string",
  "next_steps": ["string"]
}
```

Consistency rule: `verdict = GO` is incompatible with any issue of severity `CRITICAL` or `HIGH`.

3.3 CODE_PRIMER

```
{
  "structure": { "directories": "tree", "modules": "string" },
  "rules": [
    { "category": "NAMING|STRUCTURE|ERROR_HANDLING|CONCURRENCY|TESTING|DOCUMENTATION|LOGGING|
      "rule": "string", "examples": ["string"], "enforcement": "MANDATORY|RECOMMENDED" }
  ],
  "contracts": [
    { "type": "API|INTERFACE|DATA_STRUCTURE|EVENT", "definition": "string", "validation": ["s
  ],
  "anti_patterns": [
    { "pattern": "string", "why": "string", "alternative": "string" }
  ],
  "ai_rules": {
    "system_instructions": ["string"], "tool_contracts": ["string"], "interaction_logging": [
  }
}
```

3.4 TEST_STRATEGY

```
{
  "coverage": { "minimum_total": "percentage", "minimum_unit": "percentage", "minimum_integra
  "layers": [ { "layer": "UNIT|INTEGRATION|E2E|PERFORMANCE|SECURITY", "purpose": "string", "t
  "cases": [
    { "id": "string", "name": "string", "type": "UNIT|INTEGRATION|E2E", "component": "string"
      "scenario": "string", "expected_outcome": "string", "priority": "CRITICAL|HIGH|MEDIUM|L
  ],
  "doubles_and_data": [ { "name": "string", "purpose": "string", "structure": "string" } ],
  "edge_cases": [ { "case": "string", "approach": "string" } ],
  "behaviour_evaluation": {
    "applies": "true | false",
    "sets": ["identifier and version"],
    "criteria": [ { "metric": "string", "threshold": "value" } ],
    "adversarial": ["string"],
    "bias": ["string"]
  }
}
```

3.5 IMPLEMENTATION

```
{
  "files_created": [ { "path": "string", "purpose": "string", "key_functions": ["string"] } ]
```

```
  "files_modified": [ { "path": "string", "changes": "string", "reason": "string" } ],
  "dependencies_added": [ { "name": "string", "version": "string", "licence": "string", "reas
  "compliance": { "follows_primer": "true | false", "follows_plan": "true | false" },
  "next_step": "TEST_IMPLEMENTATION"
}
```

Every deviation goes in the common block `deviations`, unresolved.

3.6 TEST_IMPLEMENTATION

```
{
  "tests": [ { "file": "string", "cases": ["id"], "type": "UNIT|INTEGRATION|E2E" } ],
  "doubles_and_data": [ { "file": "string", "purpose": "string" } ],
  "execution": { "command": "string", "variables": ["string"], "requirements": ["string"] },
  "coverage_obtained": { "total": "percentage", "unit": "percentage", "integration": "percent
  "compliance": { "follows_strategy": "true | false" }
}
```

3.7 AUDIT_TESTS

```
{
  "coverage": { "actual_total": "percentage", "expected_total": "percentage", "actual_unit":
  "quality": [ { "aspect": "ASSERTIONS|EDGE_CASES|ERROR_PATHS|DOUBLES|DATA|CLARITY", "status"
  "case_review": [ { "id": "string", "implemented": "true | false", "quality": "EXCELLENT|GOO
  "verdict": "GO|GO_WITH_CHANGES|NO_GO",
  "justification": "string",
  "required_fixes": [ { "issue": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW", "recommend
}
```

3.8 AUDIT_CODE

```
{
  "fidelity_to_plan": { "score": "0-100", "deviations": [ { "deviation": "string", "justified
  "primer_compliance": { "score": "0-100", "violations": [ { "rule": "string", "location": "s
  "technical_risks": [ { "risk": "string", "category": "PERFORMANCE|SECURITY|RELIABILITY|MAIN
  "technical_debt": [ { "item": "string", "impact": "string", "recommendation": "string" } ],
  "dependencies": [ { "name": "string", "exists": "true | false", "provenance": "string", "vu
  "findings": [ { "finding": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW", "location": "s
  "verdict": "GO|GO_WITH_CHANGES|NO_GO",
  "justification": "string",
```

```
  "next_step": "VERSION|FIX_PRIMERS|PLAN"
}
```

3.9 FIX_PRIMERS

```
{
  "fixes": [
    { "id": "string", "finding": "id of the finding", "problem": "string", "file": "string",
      "change": "string", "justification": "string", "impact": "string", "regression_risk": "
  ],
  "files_modified": ["string"],
  "tests_updated": ["string"],
  "overall_regression_risk": "LOW|MEDIUM|HIGH",
  "next_step": "AUDIT_CODE"
}
```

3.10 VERSION

```
{
  "version": { "number": "MAJOR.MINOR.PATCH", "type": "MAJOR|MINOR|PATCH", "justification": "
  "changes": [ { "category": "ADDED|CHANGED|DEPRECATED|REMOVED|FIXED|SECURITY", "description"
  "breaking_changes": [ { "change": "string", "impact": "string", "migration": "string" } ],
  "migration": { "required": "true | false", "steps": ["string"], "effort": "string" },
  "deployment": { "prerequisites": ["string"], "plan": "string", "rollback": "string", "rollb
  "ai_system": { "model": "string", "parameters": "string", "instructions_version": "string",
}
```

04

Complementary cycle contracts

4.1 DOCUMENT_EXISTING (L1)

```
{
  "code_location": "string",
  "behaviour": { "purpose": "string", "functionality": ["string"], "inputs": ["string"], "out
  "components": [ { "name": "string", "type": "CLASS|FUNCTION|MODULE|SERVICE", "responsibilit
  "dependencies": { "internal": ["string"], "external": ["string"] },
  "diagrams": [ { "type": "FLOW|ARCHITECTURE|SEQUENCE", "file": "string", "description": "str
```

```
  "unverified": ["statements inferred without execution evidence"],
  "risks": ["string"]
}
```

4.2 DISCOVERY_LEGACY (L2)

```
{
  "change_objective": "string",
  "areas_affected": [ { "component": "string", "impact": "CRITICAL|HIGH|MEDIUM|LOW", "reason"
  "modification_risks": [ { "risk": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW", "mitiga
  "tests": { "existing": ["string"], "coverage": "percentage", "gaps": ["string"], "character
  "edge_cases": [ { "case": "string", "current_behaviour": "string", "has_test": "true | fals
  "recommendations": ["string"],
  "next_step": "PLAN"
}
```

4.3 DEBUG (D1)

```
{
  "incident": { "date": "ISO 8601", "description": "string", "environment": "DEVELOPMENT|STAG
  "mode": "STATIC|WITH_EXECUTION",
  "evidence": { "files": ["string"], "commands": ["string"], "logs": ["string"], "traces": ["
  "root_cause": { "summary": "string", "explanation": "string", "category": "LOGIC|CONFIGURAT
  "verdict": "CODE|CONFIGURATION|INFRASTRUCTURE",
  "actions": [ { "action": "string", "priority": "CRITICAL|HIGH|MEDIUM|LOW", "next_step": "PL
}
```

4.4 HOTFIX (U2–U7)

```
{
  "priority": "P1|P2",
  "fix_type": "TEMPORARY|DEFINITIVE",
  "root_cause": "summary of the D1 report",
  "simplified_plan": { "change": "string", "scope": "string", "risk": "string", "minimal_test
  "accelerated_review": { "verdict": "GO|GO_WITH_CHANGES|NO_GO", "justification": "string" },
  "implementation": { "files": ["string"], "changes": "string", "tests_executed": ["string"]
  "deployment": { "date": "ISO 8601", "method": "string", "emergency_approval": "person", "rc
  "monitoring": { "duration_minutes": "integer", "indicators": ["string"], "status": "STABLE|
  "post_mortem": { "timeline": ["string"], "root_cause": "string", "effectiveness": "SOLVED|M
```

```
  "definitive_plan": { "required": "true | false", "topic": "string" }
}
```

4.5 SECURITY_AUDIT (S1)

```
{
  "scope": { "files": ["string"], "components": ["string"] },
  "vulnerabilities": [
    { "id": "string", "severity": "CRITICAL|HIGH|MEDIUM|LOW",
      "category": "INJECTION|XSS|AUTHENTICATION|AUTHORISATION|CRYPTOGRAPHY|SECRETS|INPUT_VALI
      "reference": "entry in the reference list", "description": "string", "location": "strir
  ],
  "secrets": { "found": "true | false", "locations": ["string"], "recommendation": "string" }
  "vulnerable_dependencies": [ { "name": "string", "version": "string", "vulnerability": "ide
  "deployment_decision": { "critical": "integer", "high": "integer", "medium": "integer", "lo
  "proposed_residual_risks": [ { "risk": "string", "severity": "MEDIUM|LOW", "recommendation"
  "verdict": "GO|GO_WITH_CHANGES|NO_GO"
}
```

Acceptance of the residual risks is not part of the contract: a person writes it in the validation record.

4.6 BEHAVIOUR_EVAL (E)

```
{
  "set": { "id": "string", "version": "string", "cases": "integer" },
  "results": [ { "metric": "string", "value": "number", "threshold": "number", "meets": "true
  "failures": [ { "category": "string", "count": "integer", "examples": ["string"] } ],
  "adversarial": [ { "test": "string", "outcome": "PASSED|FAILED", "detail": "string" } ],
  "bias": [ { "test": "string", "segments": ["string"], "difference": "number", "threshold":
  "human_review_sample": { "cases": "integer", "selection_criterion": "string" },
  "verdict": "GO|GO_WITH_CHANGES|NO_GO",
  "justification": "string",
  "return_phase": "PLAN|IMPLEMENTATION|null"
}
```

05

Format of the contexts

The contexts are not phase artefacts and are written in Markdown with fixed sections (document 02):

```
# GLOBAL_CONTEXT (version n)
## Artefacts
## Tests
## Versioning
## Baseline security
## Data and model providers
## Prohibitions
## Parameters of the complementary cycles
## Approved models

# PROJECT_CONTEXT (version n)
## Technology
## Architecture
## Domain rules
## Compliance
## Team conventions
## Deployment
## Justified exceptions
```

06

Automatic validation

An artefact validator checks, before the human validation:

Check	Outcome if it fails
Common block present and complete, including `model`.	Incomplete artefact.
All the mandatory sections of the phase's contract.	Incomplete artefact.
Verdict, severity and priority values within the scales.	Incomplete artefact.
`verdict = GO` with open critical or high issues.	Self-approval.
`topic` in the correct format and matching the folder.	Incomplete artefact.
`iteration` consecutive with the previous artefact of the same phase.	Incomplete artefact.
Cross-references (`finding` , `case` , `issue`) existing in the previous artefact.	Incomplete artefact.

The validator **does not judge the content**: that is for the AI Reviewer and the person. A serverless tool with this function is part of the next steps of SPAD (document 08).

07

Related documents

Document	Relationship
document 01 · Operating guide	Phases that produce each artefact.
document 02 · Contexts, topics and artefact register	Model record and validation record.
document 03 · Validation policy	Causes that the automatic validator detects.
document 04 · Complementary cycles	Contracts L1, L2, D1, U and S1.
document 05 · Systems that include AI	Contract of the behaviour evaluation and `ai_system` fields.
document 06 · Instructions by phase	Instructions that require these outputs.

08

Version control

Version	Date	Changes
0.1	19-09-2026	First version. Rewrites the previous specification: common block with model record and deviations; single scales; human validation record; main cycle contracts with dependencies and licences; complementary cycle contracts; behaviour evaluation contract; format of the contexts; checks of the automatic validator.